MIL-C-52238C(ME)
13 July 1978
SUPERSEDING
MIL-C-52238B(ME)
27 August 1976

MILITARY SPECIFICATION

COOLING TOWER, LIQUID SEMITRAILER MOUNTED

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a semitrailer mounted, electric powered, counterflow, induced draft, water cooling tower with all necessary piping and auxiliary equipment (excluding electrical power source).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

W-C-375	- Circuit Breaker, Molded Case; Branch-Circuit and Service.
W-P-115	- Panel, Power Distribution.
W-R-550	- Rod, Ground (With Attachments).
CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).

FSC 3655

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FF-B-575	- Bolts, Hexagon and Square.
FF-W-92	- Washer, Flat (Plain).
GG-G-76	- Gages, Pressure and Vacuum, Dial Indicating, (for Air, Steam, Oil, Water, Ammonia and Chloro-Fluoro Hydrocarbon Gases).
QQ-A-250	- Aluminum and Aluminum Alloy Plate and Sheet; General Specification for.
QQ-B-637	- Brass, Naval: Rod, Wire, Shapes, Forgings, and Flat Products with Finished Edges (Bar, Flat Wire, and Strip).
QQ-S-781	- Strapping, Steel, and Seals.
WW-C-581	- Conduit, Metal, Rigid; and Coupling Elbow and Nipple, Electrical Conduit: Zinc-Coated.
WW-H-171	- Hangers and Supports, Pipe.
WW-P-406	- Pipe, Steel (Seamless or Welded) (For Ordinary Use).
WW-P-471	- Pipe-Fittings: Bushings, Locknuts, and Plugs; Iron, Steel, and Aluminum; (Threaded); 125-150 Pounds.
WW-P-521	- Pipe Fittings, Flange Fittings, and Flanges, Steel and Malleable Iron (Threaded and Butt-Welding) 150 Pound.
WW-U-531	- Unions, Pipe, Steel or Malleable Iron; Threaded Connection, 150 LBS. and 250 LBS.
WW-V-54	- Valve, Gate, Bronze (125, 150, and 200 Pound, Threaded Ends, Flange Ends, Solder End and Brazed Ends, For Land Use).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-0075	- Tape, Preservation and Sealing.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.

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MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-3432	- Cable and Wire, Electrical (Power and Control; Semi-Flexible, Flexible and Extra Flexible, 300 and 600 Volts).
MIL-G-3681	- Generating and Charging Plant, Liquid-Carbon-Dioxide, Portable, 300-Pound-Per-Hour Capacity.
MIL-S-4043	- Steel; Corrosion-Resisting (Extra Low Carbon Type 304), Plate, Sheet, and Strip.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-S-12514	- Starters, Motor; Across-the-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric.
MIL-S-13281	- Steel, Carbon, Alloy, and High Strength Low Alloy, for Welded Structures (Stress Relieved).
MIL-C-13909	- Conduit, Metal, Flexible; Electrical, Shielded.
MIL-V-16271	- Valves, Tank Float, Angle and Globe Pattern.
MIL-V-16721	- Valves, Foot, Low Pressure (for Water).
MIL-I-17244	- Indicators, Temperature, Direct-Reading Bimetallic, (3 and 5 Inch Dial).
MIL-F-20236	- Fittings, Tube and Pipe, Butt-Welding, 300 P.S.I. and 775° F Maximum.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-S-40626	- Sign Kit, Vehicle Class.
MIL-E-52798	- Enamel, Alkyd Camouflage.

STANDARDS

FEDERAL

FED. STD. No. 66	- Steel, Chemical Composition and Hardenability.
FED. STD. No. 595	- Colors.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

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MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-781	- Reliability Tests: Exponential Distribution.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1472	- Human Engineering Design Criteria For Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS14301	- Pipe, Steel (Seamless and Seamed) (for Ordinary Use).
MS35647-8	- Padlocks, Key Operated.
MS35844	- Plug, Machine Thread, Magnetic: Drain.
MS49006	- Plugs, Pipe, Magnetic - Headless; Aluminum, Iron or Steel.
MS75021-1	- Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28 Volt, Waterproof.

(Copies of specification and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual.

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(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York, NY 10017.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Z 35.1 - Industrial Accident Prevention Signs, Specification for.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 390 - Steel Fusion Welds, Reference Radiographs for.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. 7th Street, Miami, FL 33125.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 S. San Pedro Street, Los Angeles, CA 90013.)

COOLING TOWER INSTITUTE

CTI Bulletin ATP-105.

(Application for copies should be addressed to the Cooling Tower Institute, 4252 Richmond Avenue, Houston, TX 77027.)

DEPARTMENT OF LABOR (DOL)

29 CFR 1910 - Occupational Safety and Health Administration Standards (OSHA).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

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DEPARTMENT OF TRANSPORTATION (DOT)

49 CFR 393 - Federal Motor Carrier Safety Regulations (FMCSR).
49 CFR 567 - Certification.
49 CFR 571 - Federal Motor Vehicle Safety Standards (FMVSS).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

FORGING INDUSTRY ASSOCIATION (FIA)

Forging Industry Handbook.

(Application for copies should be addressed to the Forging Industry Association, Illuminating Building, Cleveland, OH 44101.)

FRICTION MATERIALS STANDARDS INSTITUTE, INC. (FMSI)

Automotive Data Book.

(Application for copies should be addressed to the Friction Materials Standards Institute, Inc., East 210, Route 4, Paramus, NJ 07652.)

INDUSTRIAL FASTENERS INSTITUTE (IFI)

IFI Standards.

(Application for copies should be addressed to the Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, OH 44113.)

NATIONAL ASSOCIATION OF METAL NAME PLATE MANUFACTURERS, INC. (NAME)

Metal Name Plate Industry Standards and Practices.

(Application for copies should be addressed to the National Association of Metal Name Plate Manufacturers, Inc., 1000 Vermont Avenue, NW Washington, DC 20005.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, NY 10017.)

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NATIONAL BOARD OF FIRE UNDERWRITERS

C-1 - National Electrical Code.

(Application for copies should be addressed to the National Board of Fire Underwriters, 65 John Street, New York, NY 10038.)

REGULAR COMMON CARRIER CONFERENCE (RCCC)

RCCC Recommended Practices Manual.

(Application for copies should be addressed to the Regular Common Carrier Conference, 1616 P Street, NW, Washington, DC 20036.)

RUBBER MANUFACTURERS ASSOCIATION (RMA)

Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturers Association, 444 Madison Avenue, New York, NY 10022.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.

(Application for copies should be addressed to the State of California Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95804.)

TIRE AND RIM ASSOCIATION, INC (TRA)

Tire and Rim Association Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION, INC (TTMA)

TTMA Recommended Practices.
TTMA Technical Bulletin.

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(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 2430 Pennsylvania Avenue, NW, Washington, DC 20037.)

3. REQUIREMENTS

3.1 Description. The cooling tower shall be an induced-draft, counter-flow type complete with piping and auxiliary equipment to make it a self-contained unit and trailer mounted. It shall have a minimum capacity for cooling 240 gallons of water per minute (gpm) measured by a water flow manometer through a drop in temperature from plus 115° F to plus 90° F or less using air at a dry-bulb temperature of 95° F and a wet-bulb temperature of 80° F inlet or the equivalent. This cooling requirement represents a minimum capacity of 4,000,000 Btu/hr. The cooling tower unit shall be made up of four packed cell sections, each having equal cooling capacity of a nominal 1,000,000 Btu/hr. Each section of the cooling tower shall have a minimum air flow capacity of 7500 cubic feet per minute. Provisions shall be made for operating each section individually and for operating all four sections together. All cooling tower electrical controls shall be mounted in one centrally located control panel accessible to the operator. All water control valves, where practical, shall be located in the trailer gooseneck area.

Table I. Weight and Dimensions

General Dimensions:			
(a)	DOT GVWR Per TTMA RP. No. 39	Min.	35,000 lb.
(b)	Fifth wheel weight	Max.	15,000 lb.
(c)	Axle load, (each	Max.	14,000 lb.
(d)	Overall length	Max.	396 in.
(e)	Overall height	Max.	135 in.
(f)	Overall width	Max.	96 in.
(g)	Ground clearance, min., in. (SAE J894)	Min.	9.5 in.
(h)	Ground clearance, min. (exclusive of rear suspension)	Min.	12 in.
SAE J701 Interchange Coupling Dimension:			
(i)	Landing wheel clearance, radius ("LWC")	Min.	86 in.
(j)	Swing radius, max. ("SR")	Max.	58 in.
(k)	Kingpin setting, range, ("KP")	Range	24 through 36 in.
(l)	Upper fifth wheel height		52 ± 1 in.
(m)	Angle of departure, (SAE J894)	Min.	28 degrees

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3.1.1 Weight and dimensions. The gross vehicle weight rating (GVWR) shall include the operating machinery equipment and accessories installed on the semitrailer and prepared for over the highway transport. The physical characteristics of the cooling tower shall be as specified in Table I, with the cooling tower on level ground and with the inflation pressure of the tires as recommended in the T&RA Yearbook for GVW operation at highway speed. Published component and vehicle ratings shall not be increased to meet the requirements of this specification.

3.1.2 Compatibility. The cooling tower shall be compatible for coupling and towing by any line haul, 3-axle commercial truck-tractor having a GVWR of 50,000 pounds minimum.

3.2 First article (preproduction model). The contractor shall furnish one or more cooling towers as specified (see 6.2), for examination and tests within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce cooling towers that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more cooling towers for inspection as specified in 4.4.

3.4 Material. Materials shall be as specified herein and as shown in Table II and 6.11. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4.1 Corrosion-resistant material. Where corrosion-resistant material is specified, corrosion-resistant steel SAE 300 grade series, copper, brass, bronze, silver, beryllium-copper, copper-nickel alloys, or aluminum, shall be used. For steels subject to corrosion, exclusive of steels used in semitrailer, a zinc coating of not less than 1-1/4 ounce per square foot shall be applied.

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Table II. Materials

Material	Specification	Remarks
Aluminum Plate and Sheet	QQ-A-250	
Brass	QQ-B-637	Half-Hard
Steel, Corrosion Resistant	MIL-S-4043	
Steel, Welding	MIL-S-13281	Class A

Table III. Pipe, Valves, Fittings, and Instruments

Part	Specification	Remarks
Bolts	FF-B-575	Type I, SAE J429, Grade 2
Bushings, plugs and locknuts	WW-P-471	Type III, Style B
Caps	WW-P-521	Type II
Fittings, welding	MIL-F-20236	Type I or II, Class I, wrought steel, galvanized
Gages	GG-G-76	Class I, Style A, Type I, black iron case, 6-inch, size 0 to 100 psig
Hangers	WW-H-171	Type 14
Pipe, ladder	MS14301	Extra strong
Pipe, spray	WW-P-406	Weight B, Class 2 and Weight A, Class 2
Thermometer	MIL-I-17244	Design - optional, Dial size - 3 inches, case form - as required for the application, Stem length - as required for the application, Range of degrees 20 to 240

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Table III. Pipe, Valves, Fittings, and Instruments (Cont'd)

Part	Specification	Remarks
Unions	WW-U-531	Type B, Class 1
Valves	WW-V-54	Type I, Class A
Valve, float	MIL-V-16271	Style B, Size 1-1/4 inch
Valve, foot	MIL-V-16721	4-inch size
Washers	FF-W-92	Type A, Grade I, Class A, Cadmium or zinc plated

3.5 Environmental conditions.

3.5.1 Operation. The cooling tower shall perform as specified herein in any ambient temperature from plus 120° F to minus 25° F.

3.5.2 Storage temperature requirements. The cooling tower shall not be damaged by storage under the following conditions:

Air temperature:

High - plus 160° F Time period, 4 hours. No solar radiation.
Low - minus 50° F Time period, 12 hours. No solar radiation.

3.5.3 Rain. The cooling tower shall withstand rain at a minimum rate of 3 inches per hour directed simultaneously at the roof and at all four sides for 30 minutes.

3.6 Safety.

3.6.1 Insulation, guards, and enclosures. The water cooling tower shall be in compliance with OSHA standards. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Warning signs shall be conspicuously posted on each side of the cooling towers and shall read, "CAUTION: Do not operate without blower grills in place." In addition, prominent warnings shall be provided to identify concealed hazards. Protective devices shall not impair the operating function.

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3.6.2 Vehicle safety. Additional safety characteristics of the semi-trailer portion of the cooling tower shall meet the following minimum requirements:

(a) Conform to DOT Federal Motor Vehicle Safety Standards (FMVSS), 49 CFR 567 and 571.
(b) Conform to DOT Federal Motor Carrier Safety Regulations (FMCSR), 49 CFR 393.
(c) Conform to the California Vehicle Code.
(d) Conform to the DOL OSHA requirements based on an 8-hour shift.
(e) Provide all walk step and handhold components per SAE J185.
(f) Eliminate nonfunctional sharp edges, projecting points, and excessive length of fasteners.

3.7 Human factors engineering. The characteristics of the cooling tower shall provide for operation by personnel ranging from the small man, clothed, through the large man, arctic-clothed in accordance with SAE J833. The minimum access dimensions shall conform to SAE 925. Boarding provisions shall be furnished in accordance with SAE J185 and MIL-STD-1472 ladders shall reach from the ground to the top of the work deck on each side of the cooling tower. Provision shall be made for folding and securing the lower ladder sections upward when not in use.

3.7.1 Noise level. The noise level of the cooling tower shall be in accordance with MIL-STD-1474, Category D, for steady-state noise when tested in accordance with 4.6.2.5.

3.8 Reliability. The specified Mean-Time-Between-Failure (MTBF) shall be 100 hours when the cooling tower is tested as specified in 4.6.2.6.

3.9 Maintainability.

3.9.1 Maintenance support. Assemblage or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified in 6.9 shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.6.2.7.1.

3.9.2 Maintenance ratio. The cooling tower shall have a maintenance ratio of not more than 0.06. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before and after operation checks are excluded. Maintenance shall be furnished prior to the start of any testing (see 4.6.2.7.2).

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3.9.3 Scheduled maintenance. The cooling tower shall not require scheduled maintenance more often than after each 8-hour period of operation when tested as specified in 4.6.2.6.

3.9.4 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fasteners. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.10 Performance.

3.10.1 Production capacity. The cooling tower shall cool a minimum of 240 gallons of water per minute through a drop in temperature from plus 115° F to 90° F using air as specified in 3.1 or 4,000,000 Btuh minimum. The cooling tower shall produce capacity of cooling water as specified in 3.1 within 30 minutes after startup at pressure specified in 3.11.7 while operating in the ambient temperature specified in 3.5.

3.10.1.1 Endurance. The cooling tower shall operate for not less than two 125-hour production periods when tested as specified in 4.6.2.6. During each 100-hour endurance test period a shutdown not to exceed 60 minutes duration shall be permitted once each 24 hours for performing necessary preventive maintenance lubrication, and making minor repairs such as adjusting packing glands on pump shafts and valve stems. This period shall not be counted as test time. Upon resuming operation after the shutdown period the plant shall deliver water at the specified capacity and temperature within 30 minutes after startup.

3.10.1.2 Shutdown and preparation for movement. Shutdown of the cooling tower from full production, and preparation for movement to a new location, by transporting on the highway and in cross-country travel, shall be accomplished within 1 hour by two operators.

3.10.1.3 Setup after movement. Setup and preparation of the cooling tower for operation shall be accomplished within 1 hour by two operators after the cooling tower has been transported to a new location.

3.10.2 Mobility.

3.10.2.1 Performance characteristics. The cooling tower shall conform to the transport performance characteristics specified herein when: (1) prepared for transport; (2) completely serviced and adjusted as specified

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in 3.16; and (3) towed by a commercial truck-tractor having a GVWR of at least 50,000 pounds and tandem drive axles.

3.10.2.2 Highway travel. The cooling tower shall travel over smooth, hard surfaced, primary highways at sustained speed up to 55 miles per hour (mph) minimum.

3.10.2.3 Off-road travel. The cooling tower shall negotiate improved and unimproved roads, rolling hills, and off-road terrain at speeds up to 30 mph minimum. The cooling tower shall negotiate grades up to 40 percent, traverse side slopes up to 10 percent, and shall travel through mud, sand, and snow to the intended operation sites. Without prior preparation, the cooling tower shall ford hard-bottom creek bed crossing up to 20 inches deep without damage or adverse effects due to water immersion.

3.10.2.4 Trailing and turning ability. The cooling tower shall trail the truck-tractor without weaving, sidesway, cramping or bottoming. The cooling tower shall follow the truck-tractor without exceeding the DOT FMCSR tracking deviations. Jackknife turns up to an angle of 90 degrees, as measured from straight ahead to the right and to the left, shall not cause damage to or interference between the truck-tractor and the cooling tower.

3.10.2.5 Braking ability. The cooling tower shall be in compliance with minimum braking performance DOT - FMVSS, DOT - FMCSR, and the California Vehicle Code. The brake system shall also comply with TTMA RP No. 12 and RP No. 44.

3.11 Construction. The cooling tower shall be mounted on a double-drop type center semitrailer. The cooling tower shall consist of a water reservoir (sump) having a capacity of not less than 600 gallons, four cooling water and tower sections, a water circulation pump, four cooling air circulation fans and an electrical control panel. Provisions shall be made for access to the tower compartments and reservoir for cleaning out sediment and dirt. A 1-1/2 inch pipe plug shall be provided in the tank bottom, at each corner for draining the tank. Counterweighted air inlet louvers of corrosion-resistant steel shall be furnished. The air inlet louvers shall be supported at each end by and rotate on corrosion resistant steel shafts that are supported in bearings of tetrafluoroethylene (TFE). The counterweighted louvers shall be balanced to open and close by air flow from the air circulation fans. Provisions shall be made for draining accumulations of water from all parts of the cooling tower. Drain ports and manifolds shall be of sufficient size to completely drain all water retaining vessels within 20 minutes. Metal covers shall be provided on the top of the blower casings to prevent entrance of snow and rain when the cooling tower is not in use. A means shall be provided to fasten the covers in place during

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transport. Space shall be provided for securing and storing the covers when they are not in use. The water reservoir shall be divided into four sections, one for each cooling tower section. Each section shall be baffled sufficiently to prevent loss of water while in transit. Each section of the cooling tower shall be removable from operation without affecting the operation of the three remaining sections. While any section is removed from service, its spray headers shall be removable without affecting the operation of the other three sections. Pipe, valves, fittings and instruments shall be as shown in Table III.

3.11.1 Cooling tower. Four cooling tower sections each consisting of an air blower assembly having a capacity of 7500 cubic feet per minute (CFM) with casing, packing, spray headers and nozzles, and necessary piping and fittings with a capacity to provide the cooling requirement specified in 3.1 shall be furnished. The under support mesh of each section shall be corrosion resistant steel or aluminum of sufficient strength to support the weight of the tellerettes when wet without sagging during use or while in transit.

3.11.2 Packing. The tower packing shall be 1-3/4 inches by 3/4 inch by 1/8 inch pieces of polyethylene tellerettes randomly packed in each cooler compartment to a uniform depth. The total volume of packing for the four cooling compartments shall be not less than 240 cubic feet.

3.11.3 Strainers. Two strainers constructed from corrosion resistant steel mesh shall be furnished. The strainers shall be removable for cleaning purposes. The mesh shall be of a size that will remove sediment particles 0.020 inch or larger.

3.11.3.1 Duplex strainers. A duplex strainer shall be provided in the process line for removing particles from the water supply. Provisions shall be made for removal of the strainers for cleaning.

3.11.3.2 Sediment strainer. A sediment strainer shall be provided for removing particles from the water make up line. The strainer shall be removable for cleaning.

3.11.4 Spray nozzles. Spray nozzles and headers constructed from corrosion resistant material shall be furnished. Nozzles shall be so constructed that particles not to exceed 0.025 inch cannot lodge in any nozzle in a manner that would block or restrict the flow of water through its outlet.

3.11.5 Temperature indicators. Two temperature indicators shall be provided: One to indicate the temperature of the water entering the tower, and one to indicate the temperature of the water leaving the tower.

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3.11.6 Pressure gages. Two water pressure gages shall be provided: One to indicate the water pressure entering the tower, and one to indicate the water pressure leaving the tower.

3.11.7 Water pump. A water-circulation pump, having a capacity of not less than 240 gpm at a discharge head of not less than an 80 foot head, shall be furnished. The pump shall be complete with electric motor, foot valve, with strainer, and all necessary controls.

3.11.8 Water-treatment tank. A water-treatment mixing tank having a minimum liquid capacity of 2-1/2 gallons shall be provided in the makeup water line to permit mixing water-treatment chemicals with water before adding to the tower. All necessary piping and valves shall be provided for bypassing the treatment tank when it is not in use. Provisions shall be made for retaining the water treatment material in the tank while it is dissolved into the makeup water.

3.11.9 Heater. A 20 kW electric immersion heater for operation on the 416-440 volt power supply for use in heating the reservoir water supply during startup and emergency shutdowns to prevent freezing during operation at temperatures below 32° F shall be provided. The heater shall be mounted in the tower sump. All necessary wiring and controls including an operating signal light shall be provided. A thermostatic control shall be provided to prevent the water temperature from rising above 90° F.

3.11.10 Piping.

3.11.10.1 Makeup water. A pipe of 1-1/4 inch nominal pipe size shall be provided for adding water to the sump. The pipe shall be furnished with necessary valves and piping to prime the water-circulating pump. All make-up water shall be passed through the water-treatment tank, except that a bypass line with necessary valves and fittings shall be provided to bypass the water-treatment tank. One end shall be provided with a float valve conforming to MIL-V-16271, Style B, Size 1-1/4 inch. The inlet end shall have external 1-1/4-inch NPT threads. The piping shall be braced and supported to minimize vibration or racking and shall be accessible for connecting the inlet water line.

3.11.10.2 Process line. A pipe line of a minimum of 3 inches in diameter shall be furnished. The pipe shall take the process water to the top of the towers and shall connect the pump to the duplex strainer. The inlet and discharge ends shall have external NPT threads. The normal flow of water shall be as follows: Incoming water shall enter the manifold feeding the four sections of the tower; water shall be carried from this manifold to the pipe containing the spray nozzles; water shall be sprayed from the spray nozzles and collected in the sump at the bottom of the tower; from

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the sump the water shall be drawn into the circulating pump through the foot valve and circulated through the duplex strainer to the discharge into the outlet.

3.11.11 V-belt drives. When furnished, V-belt drives shall be in accordance with Rubber Manufacturers Association Engineering Standards for Multiple V-Belt Drives.

3.12 Cooling tower electrical system. The cooling tower shall be provided with a 416 or 440 volt, 3 phase, 60 Hertz (Hz), alternating current (ac) 4-wire electrical system.

3.12.1 Wiring and fittings. Tower electric wiring, electric equipment and control installations shall be installed in accordance with NBFU, National Electrical Code Standard C-1. All wiring for power transmission to electrical components shall be Type TW, fungusproofed, AWG standard, color coded. Branch circuits for the motors shall be installed in rigid-metal conduit.

3.12.2 Power supply. A 100-foot cable conforming to MIL-C-3432, type designation CO-03HLF(3/0-3/6R)1650, shall be furnished. The load end of the cable assembly shall be provided with a 600-volt, 4-pole, polarized female plug connector with an aluminum housing for mating with and clamping to the receptacle connector. The maximum length and diameter of the plug shall be 12 inches and 4-1/4 inches, respectively. The inside diameter of the sockets shall be not less than 5/8 inch. The plug housing shall be connected to the grounding conductor and to the ground socket. The ground socket shall be identified green; no other socket shall be identified. The cable connections shall be made to the plug sockets as follows (reading *clockwise looking at the load end of the plug*):

1	White
2	Black
3	Red
4 (Ground)	Green

The other end of the cable shall be provided with bared conductors for connection to solderless terminals on the generating set. A storage area with securing devices shall be provided for storing the cable when not in use.

3.12.3 Distribution and control panel. A distribution and control panel shall be furnished for receiving electrical power from a central source and distributing it throughout the cooling tower. The panel and circuit breakers shall conform to W-P-115, Type I, Class 2, and W-C-375, Class 2d; 100 amps, 3 poles, 15,000 asym amps, 480 volts, 60 cycle, ac, type and of

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the ground fault circuit interrupter (GFI) types having electrical connection optional; the panel shall be in a weatherproof steel enclosure. A main power-supply receptacle, to mate with the plug connector (see 3.12.2), shall be provided. The receptacle shall have bolts attached with wingnuts for clamping on the cover to provide a weatherproof unit. The cover shall be attached to the housing by a chain. The receptacle housing shall be connected to the ground conductor and the ground pin. The panel shall consist of the following:

(a) A main circuit breaker to apply and to remove power from the distribution circuits.
(b) A power distribution panel with a power connection through a branch circuit breaker for each motor.
(c) A motor starter for each motor, conforming to MIL-S-12514, Type I, Class 2, Style A, 416 or 440 volt, 3-phase, size as required, horsepower as required, and full-load ampere rating as required. Starter controls shall be assembled into a control station which shall be enclosed in a NEMA Type IV (watertight) enclosure.

3.12.4 Motors. Electric motors shall be 416 or 440 volt, 3-phase, 60-cycle, totally enclosed, fan-cooled, continuous-duty, squirrel-cage induction, conforming to CC-M-1807, Type II, Class 3, Style E, Enclosure 2/K except that the frame size shall conform to NEMA Standard MG1. Motors shall be clockwise rotation and not less than 5 horsepower.

3.12.5 Grounding. All motor frames and electrical equipment enclosures shall be grounded to the semitrailer frame with bare No. 6 AWG copper conductor. The cooling tower electrical system neutrals shall be grounded to each panelboard neutral bus. A bolt with wingnut shall be mounted on the semitrailer frame in a location where the ground-rod cable can be attached by a man standing on the ground. A ground-rod shall be provided conforming to W-R-550, Type III, Class B and as follows: The ground-rod assembly shall consist of a driving bolt and three 3-foot lengths of 5/8-inch diameter rod, each with a coupling. Rods shall be made of a high-strength steel core with a copper surface bonded to the core by a fusion. Each end of each rod shall be threaded, except that one rod shall be provided with a conical point. Couplings shall be made of copper alloy. The driving bolt shall have a square end and shall be made of steel. All threads shall be 5/8-inch-11 NC. The grounding cable shall be in accordance with W-R-550, Type II. Means shall be provided for storing the ground-rod assembly and cable when they are not in use.

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3.12.6 Electrical conduit.

3.12.6.1 Rigid metal conduit. The rigid conduit shall conform to WW-C-581, size optional.

3.12.6.2 Flexible metal conduit. The flexible conduit shall conform to MIL-C-13909, size optional.

3.12.7 Phase sequence control. If reversed motor rotation will cause incorrect operation of, or damage to, any component of the cooling tower, the motor shall be equipped with a phase-sequence control to prevent reverse operation.

3.13 Semitrailer characteristics.

3.13.1 Industry standardization. The semitrailer shall conform to the applicable industry standardization as contained in the following documents:

(a) SAE Handbook.
(b) TTMA Recommended Practices and Technical Bulletins.
(c) RCCC Recommended Practices Manual.
(d) TRA Yearbook.
(e) FMSI Automotive Data Book.

3.13.2 Semitrailer specifications. The semitrailer portion of the cooling tower shall be in accordance with Table IV.

Table IV. Semitrailer Specifications

Feature	Requirement
(a) Semitrailer Body.	Double drop type; Full width front gooseneck, center drop bed and full width rear platform, all covered with metal plate or planking; gooseneck, bed, and platform suitable for installation of the cooling tower operating machinery.
(b) Semitrailer Chassis.	Provide strength, flexibility and rigidity to withstand dynamic, over-the-road, and off-road transport as well as equipment and machinery performance, while stationary at operation sites.

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Table IV. Semitrailer Specifications (Cont'd)

Feature	Requirement
(c) Frame.	Double drop, ladder or perimeter frame type; all welded construction; (HSLA) AISI high strength low alloy steel having min. 50,000 psi guaranteed yield point (YP); stress, max. 50% of YP.
(d) Intervehicular Connections.	Front airbrake and electrical Connections per RCCC RP-107; rear connections per SAE J849.
(e) Upper Coupler Assembly.	Per DOT FMCSR 393.70, TTMA RP No. 6 and 19, TOFC rated; kingpin per SAE J700, 2-in. dia., heat-treated, alloy steel type; front coupling ramp, min. 2-in. vertical height split pipe or beveled plate and gussets types; coupler plate AISI HSLA steel, same as frame; size min. 88-in. wide, cover 36-in. dia. 5th wheel; coupler plate min. 1/4-in. thick; plate and kingpin reinforcement of egg-crate type; external SAE J534 lube fittings with lines to lubricate both road and curb side of plate, when coupled.
(1) Kingpin Option.	When specified (see 6.2), heavy-duty per SAE J848, 3.5-in. dia., heat-treated, alloy steel type.
(f) Front Supports (Landing Gear).	Per TTMA RP No. 4; twin vertical lift, telescopic, non-rotating type with two-speed gearbox and folding handcrank with holder; leg travel, min. 17-in., providing upper coupler-to-ground at kingpin of at least 46-in. through 58-in.; support legs full skirt-type; both fore and aft longitudinal bracing and transverse bracing;

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Table IV. Semitrailer Specifications (Cont'd)

Feature	Requirement
	legs equipped with equalizing mud shoes having a min. 140-sq. in. area for each shoe; edges turned up; supports withstand Trailer on Flat Car operation and free-fall drop of at least 4-in. height without permanent deformation; gear boxes filled with reopectate silicone base grease intended for cold weather operations per RCCC Advisory AV7-5.
(g) Suspension System.	Tandem (two-axle) type; air-spring type intended for on-off road use; withstand both dynamic (road travel) and static (equipment operation) imposed loads and vibration; min. 8-in. travel range from compression to rebound each axle; cushioning bumper stops; antisway components linking roadside and curbside; heavy duty shock absorbers dampening ride; suspension installation in correct alinement with kingpin and semitrailer centerline per both suspension and axle mfgr. recommendations; air spring suspension of trailing member type; having piston or multiple bellows per SAE J551, internal bumpers, air pressure safety regulator valve, air controls and airtight plumbing; ride frequency max. 130-cpm at center of bed; height control valves, closed center time delay type protected from road splash, debris, and obstacle damage; time delay limits min. 3-sec. to max. 8-sec;

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Table IV. Semitrailer Specifications (Cont'd)

Feature	Requirement
	guarded driver override air control on road side front for selection of (1) maximum height, (2) normal automatic height, and (3) depleted air height; plumbing of flexible hose and reuseable swivel fittings same as for air braking system.
(h) Axles.	Two-ea. in tandem; per DOT FMVSS and TTMA RP No. 39, GAWR min. 20,000 lb. hub rating; min. 20,000 lb. brake rating and min. 25,000 lb. beam rating for each axle; hardened and ground alloy steel spindles; standardized tapered roller bearings with cups, bearings, and cones in matched sets, all from the same bearing mfgr. for each semitrailer; oil lube seals per TTMA RP No. 43 w/visual cap windows; axle hubs per SAE J694, 10 stud, 11-1/4-in. BC with outboard mounted brake drums.
(i) Tires and Wheels.	All components per TRA Yearbook and DOT FMVSS, single, disc wheel and tire heavy duty disc wheel assemblies; rim and wheel disc fabricated by same mfgr. or as recommended and approved by rim mfgr; wheels with a standard disc offset.
(1) Tires.	Widebase singles, tubeless type, min. size 18-22.5 min. LRJ (18 PR) having road lug tread design; cold inflation matched to loading per TRA Yearbook max. 65 psi; metal valve caps w/elastomer seals.

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Table IV. Semitrailer Specifications (Cont'd)

Feature	Requirement
(2) Balancing.	Tires, wheel, hub, and drum or disc, all inherently in balance and not beyond industry tolerances, to prevent wheel tramp, drumming, or excessive tire wear.
(3) Spare.	One ea. inflated spare tire and wheel assembly, furnished loose (unmounted) with each vehicle.
(j) Rear Components.	Rear platform per SAE J682 and J683; rear platform full width between frame kick-up and rear bumper. Splash guards per TTMA RP No. 41 and SAE J683; antisail type of cold-weather type reinforced plastic, polyester, or other elastomer having a min. 3000 psi ultimate strength; guards mounted to trailer brackets by metal clamping straps, fastened with hex bolts and nuts through TTMA/RCCC standard holes. Rear tow hook per SAE J849, reinforced to frame. Rear bumper per DOT FMCSR and TTMA RP No. 10; Horizontal dock bumper, min. 8-inch, height, having four ea. rubber blocks on bumper face; 2-position, retractable, hinged lower DOT bumper required, if angle of departure at normal loaded level measures less than 280 per SAE J894; retracted position prevents hang-up off-road.
(k) Brake System.	PER DOT FMVSS, DOT FMCSR, and California Vehicle Code; straight air service and emergency-parking system; withstand high speed, off-road, and mountainous travel and creek bed usage;

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Table IV. Semitrailer Specifications (Cont'd)

Feature	Requirement
	provide priority to brake system over any auxiliary air system use.
(1) Gladhands.	Per SAE J318; yellow for control (service) and red for supply (emergency), ID tags; and spring loaded automatic gladhand dust covers furnished; locations two ea. front per RCCC RP-703 and two ea. rear per SAE J849.
(2) Relay Valves.	High speed, antilock type; waterproof components.
(3) Air Reservoirs.	Min. three ea. per DOT and SAE J10 and J1070; one ea. ser., one ea. parking release and one ea. air suspension; sized per DOT and component mfgrs; automotic drain valves on each reservoir; no component lower than bottom of frame rails.
(4) 121 Antilock System.	Per DOT and RCCC RP's providing tandem axle-by-axle or brake-by-brake protection; hermetically sealed and water-immersion proof electrical components; manual reset circuit breakers providing system protection; in the sensor circuit, furnish min. 2-ea. spare fuses in same location and environmentally protect these spares; if fuses are used min. 80% of antilock system's electrical cable lengths covered or encased by grounded metal (including frame members) or system filtered to prevent external radio and electromagnetic interference; TTMA RP No. 42 indicators located near each axle; indicators of preferred RCCC RP-108 type furnished for other running lamps;

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Table IV. Semitrailer Specifications (Cont'd)

Feature	Requirement
	to reduce current draw and be compatible with the computer, twin-bulb marker lamps may have one of the two bulbs removed and a dummy plug inserted;
(5) Foundation Brakes.	Disc type or wedge or S-cam type with removable dust shields; components protected from damage, located above the bottom of the brake disc or brake drum; discs, wedges, cams, and roller pins, as a minimum, hard chrome plated and silicone high temp. grease on bearing surfaces to prevent corrosion malfunction.
(6) Brake Actuators.	Heavy duty spring type having environmental protective covers, boots, and seals; elastomer coated main spring; self-adjustors for each wheel, providing correct compensation for lining wear without lock-up or unbalance; manual means to adjust in case of malfunction.
(7) Plumbing.	All hose type per RCCC RP-305 having stainless steel reinforcement and reuseable swivel fittings permitting hose replacement from bulk hose supply; routing within protection of framing members; elastomer cushioned clamps and grommets or thru-frame connectors to secure and support hose; identify hose function adjacent to each fitting with weatherproof means; min. 3-in. of slack at relay valves.
(1) Vehicle Electrical System.	PER DOT, California Vehicle Code, SAE Handbook's Electrical Equipment and Lighting Section TTMA RP No. 9 and RP No. 14, and RCCC RP Manual;

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Table IV. Semitrailer Specifications (Cont'd)

Feature	Requirement
	heavy duty waterproof system providing protection from physical damage, shock, vibration, and environment; system 12-volt potential with negative ground per SAE J538.
(1) Front Receptacle.	Per RCCC RP-703 and similar to TTMA RP No. 6; receptacle per TTMA RP No. 40 and RCCC RP-107 receptacle with elastomer truck-tractor plug support bracket; weatherproof connection box with nut and stud terminals; SAE J553 circuit breakers protecting all live circuits, no fuses; if used, receptacle-to-remote-box cable per SAE J1067; ground wire.
(2) Rear Receptacle.	Same as front receptacle and to include dolly plug support bracket; location per SAE J849.
(3) Wiring.	Per SAE J163 and J555 and RCCC RP Manual; wire identification code per SAE J560 or, optionally, SAE J821; multiple circuits combined into harnesses; wiring supported at not less than 36-in. spacing with elastomer ties and secured mounts to prevent whipping, chafing, and mechanical or fatigue damage; interior of mechanical splicing devices waterproofed with RTV silicone sealant; RCCC RP-110 wire with RCCC RP-112 terminals; main junction blocks of nut and stud type having an electrical varnish coating after assembly; wire-to-wire terminals secured to vehicle.

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Table IV. Semitrailer Specifications (Cont'd)

Feature		Requirement
(4)	Marker Lamps.	Per RCCC RP-108/1 or RP-108/3, flush or in-line protective lens, "preferred" type; locations per RCCC RP-702 and TTMA RP No. 9 including reflectors; reflectors secured to vehicle with metal fastener(s).
(5)	Rear Combination Lamps.	PER TTMA RP No. 14 and SAE J759, SAE-IST, flush mounted elastomer housing w/bulb and snap-on or elastomer gasketed lens or sealed unit with elastomer ring; round 4-in. dia. size installed on TTMA standard 4-1/2 in. hole; double filament configuration with 3-wire pigtail, including ground; waterproof and shock resistant; total four ea., outboard two ea. turn and clearance and inboard two ea., stop and tail functions.
(6)	Wiring Diagrams.	Metal plate showing wiring furnished per TTMA Technical Bulletin No. 65 and "Metal Name Plate Industry Standards and Practices" NAME.

3.14 Military vehicle electrical system. When specified (see 6.2), the cooling tower shall be equipped with a military compatible 24-volt service lighting system to provide towing capability for the military design trucks as well as commercial trucks. In addition to the TTMA 7-contact receptacle, the trailer shall be equipped with a military 24-volt, 12-contact inter-vehicle receptacle in accordance with MS75021-1, having a spring-loaded, waterproof cover. The military receptacle shall be located near the TTMA receptacle and both shall be labeled with identification plates for 24-volts and 12-volts respectively, negative ground. The military 24-volt wiring shall be routed into a weatherproof stud and nut type junction box, equipped with SAE J553 circuit breakers, protecting live circuits B, E, and J of the military service lamp portion. All unused leads of the cable shall be insulated and sealed watertight. The military 24-volt circuits shall be connected through replaceable resistors, for live circuits, and to the trailer frame, for ground circuits, in accordance with Table V.

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Table V. Military-Commercial Connections

Function	Mil. Receptacle Terminal	Connection Instructions	Comm. TTMA Wiring Ident.
Blackout Marker	A	No.	-
Ser. Left Turn & Stop	B	Connect to -	YEL (3)
Blackout Marker	C	No.	-
Ground	D	Connect to -	WHT (1)
Ser. Tail & Marker	E	Connect to -	BLK (2)
		& Connect to -	BRN (6)
Blackout Stop	F	No.	-
Blackout Marker	H	No.	-
Ser. Right Turn & Stop	J	Connect to -	GRN (5)
Aux. Power (Hot)	K	No.	BLU (7)
Ground	L	Ground to Frame	-
Electric Brake	M	No.	-
Electric Brake	N	No.	-

With the military system installed, the TTMA marker circuits, color coded BLK (2) and BRN (6), shall be electrically isolated from each other, thus preventing 12-volt current cross-over when operated by commercial trucks through the 12-volt TTMA plug. Replaceable diodes may be used, having a minimum amperage of 150 percent of the current demand of the running lamps served. Military circuits letter coded B and J on the tactical trucks are passenger car combination stop and turn signal type, using the same lamp filament, and there is no single stop lamp circuit. As a result, the cooling tower's turn signal lamp circuits, YEL (3) and GRN (5), shall function as both turn signal lamps and stop lamps and the cooling tower's normal stop circuit, RED (4), cannot be energized from the military 24-volt system input.

3.15 Electromagnetic interference. The electromagnetic interference emission of the cooling tower shall conform to MIL-STD-461, Notice 4, para. 4.1.1.2.

3.16 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating. The cooling tower shall be equipped with a lubrication plate conforming to SAE J753, located near the gooseneck.

3.16.1 Lubricants. The cooling tower shall operate as specified herein when lubricated with military lubricants (see 6.4).

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3.16.2 Lubrication fittings. Lubrication fittings shall conform to SAE J534, threaded types. Fittings shall be located in a protected position and shall be accessible to a commercial 15 ounce grease gun. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.16.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment will damage grease seals or other parts.

3.16.4 Filling, draining, and checking provisions. Enclosures such as gear cases and transmission housing that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a drain plug conforming to MS35844 or MS49006. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be to the ground when the cooling tower is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet to the ground.

3.16.5 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F.
(b) MIL-G-10924 - For bearings where operating temperatures do not exceed 225° F.

The cooling tower shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.16.6 Oil lubrication. Unless otherwise preserved for packaging (see 5.1.1), all components shall be filled to the operating level with winter grades of military oils designated for use in the temperature range of 0° F to 32° F. Each component shall be tagged in a conspicuous place to indicate the temperature range and grade of lubricant used.

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3.16.7 Servicing and adjustment. Prior to delivery, the contractor shall service and adjust each semitrailer for customer operational use, including at least the following:

(a) Adjustment of operating components.
(b) Flushing and cleaning of the hydraulic system, air brake system and air suspension system.
(c) Adjustment of the brake system.
(d) Adjustment of the suspension and axles.
(e) Continuity inspection of the vehicular electrical system.
(f) Check torque of all wheel lugs and inflation of all tires.
(g) Complete lubrication.

3.17 Fungus and moisture resistance. The electrical circuitry of all electrical systems, including all components and connections except as specified below, shall be protected from the effects of moisture and fungus growth by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.17.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements which are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component and element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.17.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.18 Slinging provisions. Permanently affixed slinging provisions that enable the cooling tower to be lifted in its normal travel position shall be provided and shall conform to MIL-STD-209. Slinging provisions may also be used as tiedowns when such provisions conform to MIL-STD-209.

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3.19 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the cooling tower to the floor or deck of a transport medium shall be provided and shall conform to MIL-STD-209. Tiedown provisions may also be used as slinging provisions when such provisions conform to MIL-STD-209.

3.20 Identification marking. Marking shall be applied to the cooling tower in the area of the operator's station on metal plates. The plates shall be in accordance with the "Metal Name Plate Industry Standards and Practices", as published by the National Association of Metal Name Plate Manufacturers (NAME). Identification and shipping data plates shall be of material in accordance with MIL-P-514, Composition C of Type I, Grade A, Class 1 material. The plates shall be secured to the cooling tower with metal fasteners in accordance with the NAME Standards and Practices.

3.20.1 Identification plates. The cooling tower shall be identified in accordance with the Federal Department of Transportation, the manufacturers, and MIL-STD-130 or MIL-P-514, Type I, Style I plates.

3.20.2 Shipping data plates. The cooling tower shall have shipping data applied to MIL-P-514, Type III plates. These plates shall show the silhouette of the cooling tower in the transport position, indicating dimensions at the center of gravity (CG) and the location and capacity of the lifting and tiedown attachments. The plates shall be fastened in a conspicuous protected location on the curbside of the cooling tower.

3.20.3 Instruction plates. The cooling tower shall be equipped with instruction or pictorial plates describing any special or important procedures to be followed in assembling, operating, or servicing. The safety, warning, caution, and notice plates shall be in accordance with ANSI Z 35.1. Plates shall be attached in a conspicuous protected location.

3.20.3.1 Flow diagrams. A flow diagram illustrating each process of the cooling tower, and a flow diagram illustrating the overall process of the cooling tower shall be permanently affixed near the controls. The flow diagrams shall be of a size to be clearly readable.

3.20.3.2 Wiring diagram plates. Electrical wiring schematic diagrams for the nonvehicular wiring circuits shall be affixed near the controls. The plates shall be of a size to be clearly readable.

3.20.3.3 Operating instructions. Operating instructions shall be furnished and shall be permanently affixed in a readable location near the controls. Instructions shall be clear and concise and shall contain sufficient information to enable operation without damage to the cooling tower or injury to personnel.

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3.20.4 Valve marking. All control valves and safety valves shall be identified by a permanently affixed plate on or near the valve indicating a flow control number. Composition of the plate shall be such that its legibility will not be impaired by exposure to oil, dirt, and light. The use of tags will not be acceptable.

3.20.5 Vehicle marking. The registration numbers and other markings of the cooling tower shall be in accordance with MIL-STD-642 (see 6.5). The GVW of the cooling tower shall be marked on each side of the cooling tower gooseneck in a manner to be discernible. The prescribed tire pressures shall be marked on the side of the semitrailer near each tire. Unless otherwise specified (see 6.2), all marking shall be lusterless white, Color No. 37875 conforming to FED. STD. No. 595.

3.20.6 Treatment and painting. The portions of the cooling tower and its components and parts normally painted in accordance with the contractor's commercial practice shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A. The inaccessible areas of the cooling tower shall be treated with antirust coating material in accordance with TTMA Technical Bulletin No. 56. The exterior finish of the cooling tower shall be Color, Forrest green, in accordance with MIL-E-52798.

3.21 Contact surfaces of dissimilar metals. Contact between dissimlar metals shall be avoided when practical. The definition of dissimilar metals shall conform to MIL-STD-889. When such contacts do occur, they shall be separated by the use of joint fillers composed of zinc chromate, nonconductive aluminum powder, inert filler, oils, and thinner, and by the use of membrane materials. Membrane materials shall be of elastomer stripping not less than 1/32-inch thickness. Steel parts, including bolts, nuts, screws, and washers, shall be zinc or cadmium coated in addition to the above treatment.

3.22 Government-furnished property. When specified (see 6.2 and 6.7), the following property in the quantities indicated will be furnished by the Government:

Item No.	Description	Identification	Quantity for each cooling tower
1	Padlock w/two keys	NSN 5340-00-682-1504	1 each
2	Vehicle classification sign kit	NSN 9905-00-569-6267	1 kit

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3.23 Transportability.

3.23.1 Rail. Unless otherwise specified (see 6.2), the unpackaged cooling tower shall withstand rail humping without damage or permanent deformation to any part or component when tested as specified in 4.6.2.8. After testing, the cooling tower shall operate as specified herein for a period of not less than 1 hour without malfunction or permanent deformation of any part of the cooling tower.

3.24 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.25 Vehicle weight classification. The contractor shall complete the Government-furnished vehicle weight classification forms and apply the vehicle weight classification number(s) to the cooling tower as requested (see 6.6). The vehicle classification sign kit conforms to MIL-S-40626.

3.26 Toolbox. The cooling tower shall be provided with a weatherproof toolbox made of steel not less than 0.07 inch thick. The toolbox shall have an overlapping lid mounted on continuous piano hinges with a 0.125 inch diameter pin. When the toolbox is mounted on the cooling tower, the toolbox lid shall open to at least 90 degrees from the closed position and remain there either by its own weight or by holding devices. Two hook-type fasteners shall be provided to keep the lid closed. The fasteners shall be attached to the toolbox by either bolts or rivets. A hasp which will accommodate a lock conforming to MS35647-8 shall be provided for locking. The toolbox shall be fastened to the cooling tower in a protected position. Drain holes shall be provided in the bottom of the toolbox. The toolbox shall be of a size to hold the operating and maintenance tools and repair parts as specified. Padlocks will be furnished by the Government (see 3.22).

3.27 Workmanship.

3.27.1 Metal fabrication. Metals used in the fabrication of the cooling tower shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting of ferrous metals may be employed instead of shearing or sawing. Burned surfaces of flame cut material shall

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be ground or machined sufficiently to remove ash and cooling checks. Aluminum alloys up to 1/2 inch in thickness may be cut by shearing; above 1/2 inch in thickness, the alloys shall be cut by common or high-speed saws. All bends of a major character shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.27.2 Welders, welder operators, and welding.

3.27.2.1 Welders and welder operators. Before assigning any welder or welder operator to welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by any of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedure of the American Welding Society.

3.27.2.2 Metal welding. All metal welding shall be in accordance with the AWS code and as specified herein. Welds shall provide the maximum notch toughness and fatigue strength with a minimum of residual stresses in accordance with the welding code. All welds made on high-stress members shall be made from core or heavy-coated electrodes, gas-shielded, or submerged arc. Electroslag and electrogas processes, when available, shall be included for use on heavy sections, 1 inch or more in thickness. The design of welded connections shall be in accordance with "Section 2 Design of Welded Connections" of the "AWS D1.1 Structural Welding Code". Welds which directly transverse high-stress members, and spot tacks, or intermittent welds for strength are not acceptable. Deficiencies in workmanship shall not be covered up by using filler materials. Fillet welds shall be provided, when necessary, to reduce stress concentrations. The surfaces to be welded shall be free from scale, paint, grease, and other foreign matter. When making multi-pass welds, cleaning shall be accomplished after each pass and before another layer of weld is deposited. All welded parts and assemblies shall be free from cracks and other discontinuities that may reduce the effectiveness of the part or assembly. After welding, all weldments shall be cleaned to remove slag, flux, and weld spatter. Except for vertical welding of the electroslag process, the work shall be positioned for flat welding whenever practicable. No undercuts or overlays will be permitted on any weldment. Where different metals are welded to each other, filler metal having strength of the lower strength material shall be used. When the same metals, or metals having similar properties are used, the deposited filler metal shall develop the full base metal strength of the particular weld joint or weldment. Mechanical properties of all

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weldments shall be such that they will transmit stress without weld failure or permanent deformation when subjected to any test specified in this specification. Weld joints subjected to stresses exceeding 50,000 psi shall be inspected for soundness by radiography. The standard of quality shall be that determined by the "American Society for Testing and Materials E390 Reference Radiographs for Steel Fusion Welds".

3.27.2.3 Aluminum welding. Aluminum welding shall be accomplished by the gas-metal-arc or the gas-tungsten-arc method or any other method that will provide equivalent mechanical properties of the filler metal to base metal combinations. When machines are available, all aluminum welds shall be made by automatic machines. All welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds. Aluminum welds may be ground, filed, wire brushed or chipped; hammered welds shall not be acceptable. Preheating for welding is permissible, providing the temperature does not exceed 400° F for a total time of 30 minutes. Aluminum weld processes requiring the use of welding flux shall not be used.

3.27.3 Brazing. Brazing of aluminum and aluminum alloys shall be performed in accordance with the nationally recognized and accepted recommended procedures and standards of the aluminum manufacturing industry.

3.27.4 Fiberglass fabrication. Fiberglass reinforced plastic (FRP) and FRP coated plywood used in fabrication of the cooling tower shall be smooth surfaced and free from bubbles, glazing cracks, and discontinuities and shall have a uniform FRP thickness. All edges and openings of FRP coated plywood shall be completely sealed to prevent water intrusion between the laminates. The FRP shall be free of parting agents which impair paint adhesion. Welding, bonding, connections, and fabrication of the FRP shall be in accordance with the recommended methods of the manufacturer of the material for this application.

3.27.5 Bolted connections. Boltholes shall be accurately punched or drilled and shall have burrs removed. Integral or separate washers or lockwashers shall be provided. Self-locking nuts are acceptable in lieu of standard nuts and lockwashers. All capscrews, bolts, and nuts shall be tight and correctly torqued.

3.27.5.1 Critical bolted connections. Critical load-carrying bolted connections shall, in addition to the above requirements, be as specified herein. Boltholes shall be reamed as required. All critical junctures and joints shall be fastened with high-strength bolts with washers and nuts, hex or 12-point type. Bolts shall be not less than SAE J429, Grade 8, in accordance with IFI Standards 103 and 111. Integral flange fasteners

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or hardened steel flat washers shall be furnished and sized to spread the clamp loads such that brinnelling or embedding in the component members shall be prevented. Mating nuts shall be of the prevailing-torque type in accordance with IFI Standards 100, 101, 107, and 108 minimum, Grade C. To insure constant and correct torque performance, fastener installation accuracy shall be achieved by the turn-of-the-fastener method.

3.27.6 Riveted connections. Riveted connections shall be in accordance with SAE J492. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.27.7 Aluminum bolt and rivet holes. Bolt and rivet holes in aluminum alloys shall be made by any one of the following methods:

(a) Holes may be subpunched or subdrilled undersize and reamed to finish size upon assembly.
(b) Holes may be drilled to finish size at assembly.
(c) Holes may be drilled to finish size prior to assembly with accurately mated steel drilling jigs. The jigs shall be equipped with hardened steel bushings to insure the correct alinement and location of the holes.

With all the above methods, all burrs and rough edges of the holes shall be removed prior to final assembly.

3.27.8 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for its intended use.

3.27.9 Forgings. Forged components shall be in accordance with the Forging Industry Association Handbook.

3.27.10 Machine work. All machined parts shall be manufactured to fabricated gages through the use of correct jigs and fixtures that will assure the interchangebility of all assemblies and parts. Finished contact and bearing surfaces shall be true and exact.

3.27.11 Cleanliness of hydraulic and pneumatic systems. The hydraulic and pneumatic systems of the cooling tower shall be purged to remove any foreign material after the appropriate assembly phase is completed. The sequence of purging shall be first through the lines and then second through both the lines and the components.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. Failure of any test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction cooling tower shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction cooling tower shall be subjected to the tests marked "X" in Column 1 of Table VI. Failure of any test shall be cause for rejection.

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4.4 Initial production inspection. When specified (see 3.3), one or more initial production cooling towers will be selected at random by the Government from cooling towers being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in Column 2 of Table VI to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production cooling tower shall not exclude the remaining cooling towers from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production tests, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production cooling tower to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production cooling tower and shall be cause for refusal by the Government to continue acceptance of production cooling towers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on cooling towers previously accepted and produced under the contract. Any deficiencies found as a result of the initial production testing will be considered prima facie evidence that all cooling towers accepted prior to the completion of initial production testing are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each cooling tower shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Test. Each cooling tower shall be subjected to the tests marked "X" in Column 3 of Table VI. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The cooling tower shall be examined for the following defects:

101. Dimensions and weight not as specified.
102. Parts or components missing.

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103. Assembly incorrect or incomplete.
104. Parts or components do not function.
105. Parts or components not accessible.
106. Safety devices or guards missing or not as specified.
107. Materials not as specified.
108. Materials at joints subject to galvanic action.
109. Joints, seams, or edges do not fit; misalined.
110. Treatment and painting not as specified.
111. Fasteners or locking devices incomplete, missing, or not as specified.
112. Components dirty.
113. Welding not as specified.
114. Lubrication fittings not as specified.
115. Storage space for electrical and other accessories not provided.
116. Identification markings, operation instructions, or technical publications missing or not as specified.
117. Ground clearance not as specified.
118. Corrosion protection not as specified.
119. Workmanship not as specified.
120. Threaded parts not in accordance with Handbook H28.
121. Common parts not utilized.
122. Electrical systems or components not as specified.
123. Piping not as specified.
124. Semitrailer components not as specified.
125. Lubrication provisions not as specified.
126. Toolbox not as specified.
127. Electromagnetic interference suppression components and network not identical with approved model.
128. Stress analysis for the cooling tower not furnished, does not include load, shear, and moment calculations and diagrams, produces a stress greater than 50 percent of the material yield point, or contains errors.

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Table VI. Test Schedule

Preproduction	Initial Production	Quality Conformance Individual	Inspection Comparison	Tests	Test Para.	Requirement Paragraph
1	2	3	4	5	6	7
X	X	X	X	Test conditions.	4.6.2.1	Section 3
X	-	-	-	Preassembly.	4.6.2.2	Section 3
X	-	-	X	Slinging provisions.	4.6.2.4	3.18
X	-	-	X	Tiedown provisions.	4.6.2.3	3.19
-	X	-	-	Reliability.	4.6.2.6	3.8
X	X	-	-	Maintainability.	4.6.2.7	3.9
X	X	-	X	Safety.	4.6.2.16	3.6
X	-	-	X	Noise level measurement.	4.6.2.5	3.7.1
X	-	-	X	Railroad car impact.	4.6.2.8	3.23.1
X	-	-	X	Mobility.	4.6.2.9	3.10.2
X	-	-	X	Temperature.	4.6.2.10	3.5
X	-	-	X	Rain.	4.6.2.11	3.5.3
X	-	-	X	Electromagnetic interference.	4.6.2.12	3.15
X	X	-	-	Human factors.	4.6.2.13	3.7
X	-	-	X	Performance.	4.6.2.14	3.10, Section 3
-	-	X	-	Operation.	4.6.2.15	Section 3

4.6.2 Tests.

4.6.2.1 Test conditions. Prior to testing, the cooling tower shall be lubricated with oils and greases as specified in 3.16 except that oils shall be those designated for use in the ambient air temperature at the place of test. Operational tests shall be performed with the cooling tower powered by the electric motor from a commercial or military electric power source. The cooling tower shall operate as specified herein without maintenance

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other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.6.2.2 Preassembly.

4.6.2.2.1 Examination of welds. Welds shall be examined for defects as defined in AWS Structural Welding Code. Defective welds shall be corrected as required by the AWS Structural Welding Code and reinspected. Any weldment containing over 10 percent localized weldment defects shall be replaced in its entirety. Where surface examination reveals questionable conditions, radiographic analysis shall be employed to determine adequacy of weld.

4.6.2.3 Tiedown provisions.

4.6.2.3.1 Test procedure. With the cooling tower anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209 to each provision. Measure the dimensions on the tiedowns.

4.6.2.3.2 Failure criteria. Weld failures, permanent deformation of the cooling tower or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.6.2.4 Slinging provisions.

4.6.2.4.1 Test procedure.

(a) Lift the cooling tower and hold it in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the cooling tower. Determine the forward angles of application for each provisions. Measure the clearance between each sling and the cooling tower. Measure the eye openings and clearance dimensions of the provisions.

(b) Restrain the cooling tower by anchoring the main frame and subject each slinging provision to a force of 2-1/2 times the load it carried when suspended. This force shall be applied in the direction as determined in (a), above. An acceptable alternate method of test shall be to lift the cooling tower as in (a), and add weights to the cooling tower until the force on each provision is 2-1/2 times the initial force. Hold the cooling tower under either of the above loading conditions for not less than 90 seconds.

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4.6.2.4.2 Failure criteria. Weld failure, permanent deformation of the cooling tower or any slinging provisions, slings converging more than 24 feet above the lowest extremity of the cooling tower, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.6.2.5 Noise level measurement.

4.6.2.5.1 Operator's area. Measure the noise level at the operator's station (gooseneck) in accordance with MIL-STD-1474 for steady-state noise while operating at full capacity with the cooling tower located on a paved surface. Nonconformance to 3.7.1 shall constitute failure of this test.

4.6.2.5.2 Exterior spectator position. Measure the exterior sound level in accordance with MIL-STD-1474 for exterior noise. Nonconformance to 3.7.1 shall constitute failure of this test.

4.6.2.6 Reliability. Using the MTBF specified in 3.8, the cooling tower shall be tested as specified in 4.6.2.14 with "accept" and "reject" criteria in accordance with MIL-STD-781, Test Plan IV. Continue testing until either an "accept" or "reject" decision is reached.

4.6.2.6.1 Failure definition. A failure is defined as any malfunction which cannot be corrected within 30 minutes, (all waiting periods are excluded), by adjustments, repairs, or replacement action using controls and onboard equipment tools or parts available to the using element and is a malfunction which causes or may cause one or more of the failure conditions indicated herein.

4.6.2.6.2 Test procedure. Test procedure shall be in accordance with the consecutive tests included in 4.6.2.9 and 4.6.2.14 with the operating time specified in 4.6.2.14 extended as necessary.

4.6.2.6.3 Failure criteria. Nonconformance to 3.8 shall constitute failure of this test.

4.6.2.7 Maintainability.

4.6.2.7.1 Maintenance support evaluation. Maintenance operations required during initial production testing shall be accomplished to determine conformance to 3.9.1. Failure of the support elements to maintain the cooling tower as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or MTBF.

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4.6.2.7.2 Maintenance ratio. The maintenance ratio shall be computed during initial production testing. Nonconformance to 3.9.2 shall constitute failure of this test.

4.6.2.8 Railroad car impact.

4.6.2.8.1 Test procedure. Anchor the unpackaged cooling tower on a domestic service railway freight car (blocking is permitted). Hump the car with the cooling tower tied down at speeds of not less than 4, 6, and 8 mph respectively, against five empty freight cars with their brakes set and draft extended. The cooling tower shall face in the direction of the five cars. Repeat the procedure at not less than 8 mph with the rear of the cooling tower in the direction of the five cars. The cooling tower may be reversed by either reversing the direction of the flatcar (preferred), or by reloading the cooling tower in the proper direction on the freight car. Once the test has begun, there shall be no readjustment of load, nor any reconditioning of the blocks, choking material, or tiedowns except as necessary to reverse the direction of the cooling tower. After each impact, the cooling tower shall be inspected. After completion of the railroad car impact test, an operational checkout of the cooling tower shall be conducted. Nonconformance to 3.23.1 shall constitute failure of this test.

4.6.2.9 Mobility. With the cooling tower prepared for travel, complete with all equipment installed and secured, couple the cooling tower to the applicable truck-tractor and perform the mobility tests as shown in Table VII and as follows:

Table VII. Road Test

Course	Mileage and speed
Highway	60 mi. at up to 55 mph, min.
Secondary roads	60 mi. at up to 30 mph, min.
Off-road	30 mi. at up to 20 mph, min.

During the road tests perform the following and record actual results:

(a) Tow the semitrailer over minimum 7-inch-high obstacles such that all combinations of semitrailer wheels have passed over the obstacles. Observe tire-to-surface contact, evidence of binding or interference, and, after returning the suspension to normal, measure the diagonals and parallels. Questionable tire-to-surface contact shall be retested with single sheets

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of newspaper under the tires. Ability to remove the newspaper from under any tire shall be evidence of insufficient tire-to-surface contact.

(b) Tow the semitrailer through mud at least 12 inches deep for not less than 100 feet. Any prime mover may be used for this phase.

(c) Tow the semitrailer over V-ditches, negotiating the ditches both perpendicular and at 45 degrees to direction of travel for at least 20 times, total. V-ditches shall be the standard highway type of at least 36 inches wide and 18 inches deep.

(d) Measure and record the deviation or off-tracking of the semi-trailer with respect to the truck-tractor, when towed straight ahead on a level-paved-surface, and compare with DOT FMCSR.

(e) Jackknife the semitrailer at angles of at least 90 degrees from straight ahead both to the left and to the right.

(f) Turn the combination in the sharpest rolling turn to the right for two complete circles and repeat in the opposite direction to the left.

(g) Demonstrate the brake performance of the semitrailer for conformance to DOT FMVSS, DOT FMCSR, and the California Vehicle Code.

(h) Test the semitrailer air brake system as specified in TTMA RP No. 12 and RP No. 44.

(i) Demonstrate compatability of the semitrailer to couple and uncouple from truck-tractors. Demonstrate the ability of the front supports to raise and lower the semitrailer and to withstand free falls, during uncoupling, by rapidly pulling the truck-tractor out from under the semitrailer.

Failure to complete the mobility test, inability to function and operate as specified, evidence of damage or permanent deformation or nonconformance to industry standards or legal requirements shall constitute failure.

4.6.2.10 Temperature. The empty cooling tower shall be exposed to the temperature specified in 3.5.1 and 3.5.2 for the periods specified. The temperature shall be increased to plus 32° F and plus 120° F and operated for the periods specified. Any evidence of damage, leakage, malfunction of controls, or pressure not within specified limits shall constitute failure of this test.

4.6.2.11 Rain. The complete cooling tower with all machinery and mounted equipment, shall be subjected to a simulated rainstorm while prepared for storage and during normal operation. The apparatus for producing the spray shall emit the water in the form of droplets rather than as a fine mist at

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the rate per minute and in the direction specified in 3.5.3. Water leakage through any inclosure or into any electrical apparatus shall constitute failure of this test.

4.6.2.12 Electromagnetic interference. The cooling tower shall be subjected to the tests specified in MIL-STD-461, Notice 4, Para. 4.1.1.2. The contractor shall furnish the contracting officer with the test report for evaluation within the time frame specified (see 6.2). Disapproval of the report shall constitute failure of this test.

4.6.2.13 Human factors. The cooling tower shall be evaluated to determine the human factors criteria during testing as specified in 4.6.2.6. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.14 Performance test. The cooling tower shall be operated continuously at the heat load specified in 3.1 and 3.10.1 for a period of 24 hours after stabilization of cooling level. The tests shall be conducted in accordance with the procedures outlined in the Cooling Tower Institute (CTI) Bulletin ATP-105-Parts I and II. The following readings shall be taken and recorded at the beginning of the test and at the end of each hour of operations: The temperature (wet and dry bulb) of the ambient air at the entrance, the exit, and at the fan discharge; amount of waterflow and the temperature of the water at the inlet and discharge point of the cooling tower. The hourly readings shall be totaled at the end of each 4-hour period then averaged to determine the hourly rate. The amount of water that passes through the cooling tower during the test shall be calculated and multiplied by the temperature difference to obtain Btu/hr value. This result shall be revised on the basis of the design air temperature and humidity. This projected performance should be at least the nominal rating of 4,000,000 Btu/hr of cooling. Failure of the cooling tower to remove 4,000,000 Btu/hr from the process water or failure of any component shall constitute failure of this test.

4.6.2.15 Operation. Each cooling tower shall be operated without heat load for a period of 4 hours to determine the mechanical operability of the cooling tower. For purposes of this test, the water pumped from the tower pump may be recirculated directly back to the tower. The mobility test shall be the same for each cooling tower except that the towing distance shall be 30 miles over unimproved roads. Inability to operate for 4 hours or be towed for 30 miles without repair, maintenance, change or alteration shall constitute failure of this test.

4.6.2.16 Safety evaluation. The cooling tower shall be visually inspected on receipt and shall be evaluated throughout testing for compliance with safety requirements of 3.6. Nonconformance to 3.6 shall constitute failure of this test.

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4.7 Inspection comparison. The Government may select cooling towers at any time during the contract production period and subject these cooling towers to the tests marked "X" in Column 4 of Table VI to determine that the quality of the selected cooling tower is equal to the quality standards established during the initial production inspection. The inspection will be performed by the Government at a site selected by the Government. Cooling towers will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production cooling tower.

4.7.1 Inspection failure. Failure of any cooling tower when inspected as specified in 4.7 may be considered cause for refusal by the Government to continue acceptance of cooling towers until objective evidence furnished by the contractor reveals that corrective action has been taken to eliminate the condition that caused the rejection.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a completely processed cooling tower prepared for shipment shall be considered a unit of product.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

129.	Materials, methods, and containers not as specified for Level A. Each incorrect material, method, or container shall constitute one defect.
130.	Unprotected metal surfaces not coated with preservative as specified for Level A.
131.	Semitrailer not preserved in accordance with the referenced document for Level A.
132.	Water system not drained and valves positioned as specified for Level A.
133.	Gage and thermometers not protected as specified for Level A.
134.	Float not cushioned and secured as specified for Level A.
135.	Openings into electrical components not sealed as specified for Level A.
136.	Power supply cable not coiled and secured as specified for Level A.

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137. Maintenance tools not preserved in accordance with the referenced document for Level A.
138. Consolidation not as specified for Level A.
139. Consolidation boxes not positioned and secured as specified for Level A.
140. Marking missing, illegible, incorrect, or incomplete for Level A or Commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unprotected surfaces. Unpainted exterior surfaces of items such as adjusting rods, piping, springs, linkages, grease fittings, bolts, nuts, pins and washers or any other exterior metal surfaces not specified herein requiring a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.2 Semitrailer. The components of the semitrailer chassis and body shall be preserved in accordance with the Level A (mobile) requirements of MIL-STD-281.

5.1.1.3 Water system. All components of the water cooling system such as the pump piping, valves, strainers, and tank shall be drained and dried with dry compressed air. All drain valves except the main sump drain valve, shall be closed. No contact preservative shall be applied to the interior of the water system components.

5.1.1.4 Gages and thermometers. The glass surfaces of all gages and thermometers shall be protected with a fitted piece of plywood or hard board secured in place with tape conforming to PPP-T-0075, Type II.

5.1.1.5 Float valve. The float shall be cushioned and secured in the tank to prevent movement and damage to the float and valve mechanism.

5.1.1.6 Electrical components. Openings into junction boxes, receptacles, and similar fixtures that will permit the entrance of dirt or water shall be covered with tape conforming to PPP-T-0075, Type II.

5.1.1.7 Power supply cable. The power supply cable shall be coiled to not less than 10 times its diameter and the coil secured with not less than three equally spaced ties of cotton tape or twine.

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5.1.1.8 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.1.1.9 Maintenance tools. Maintenance tools shall be preserved in accordance with the Level A requirements of PPP-P-40.

5.1.1.10 Technical publications. Technical publications shall be preserved together in accordance with MIL-P-116, Method IC-1, or IC-3.

5.1.1.11 Consolidation. The power cable, repair parts, maintenance tools, and technical publications shall be consolidated in the toolbox when space permits. When there is not sufficient space in the toolbox, the additional items shall be consolidated in a close-fitting box conforming to PPP-B-601, Overseas Type, style optional. The toolbox lid shall be secured in a manner to prevent pilferage. The wood box shall be closed and secured as specified in the appendix to the box specification. Strapping shall not be required.

5.1.2 Commercial. Commercial preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A. The semitrailer mounted cooling tower shall be prepared for shipment by towaway. The consolidated container shall be positioned on and secured to the equipment by strapping or bolting, in a manner so as not to increase cubage or interfere with lifting or towing the semitrailer. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, or Class 2, Type VI, and unless otherwise specified shall be Finish B (see 6.2). When specified Finish A strapping shall be used.

5.2.2 Commercial. Commercial packing of the semitrailer mounted cooling tower shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

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6. NOTES

6.1 Intended use. The cooling tower is intended for use in removing heat from process water used by a carbon-dixoide generating plant covered by MIL-G-3681.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of preproduction model and number of preproduction cooling towers required (see 3.2).
(c) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(d) When initial production testing is required and number of initial production cooling towers required when applicable (see 3.3).
(e) When the heavy duty SAE J848 kingpin is specified (see 3.13.2).
(f) When a military 24 volt electrical system receptacle is required (see 3.14).
(g) When lusterless white vehicle marking is not required (see 3.20.5).
(h) When Government-furnished property is required (see 3.22 and 6.7).
(i) When the cooling tower is not subject to rail humping (see 3.23.1).
(j) Time interval required for submission of electromagnetic interference report (see 4.6.2.12).
(k) Degree of preservation and packing required (see 5.1 and 5.2).
(l) When Finish A strapping is required (see 5.2.1).

6.3 Preproduction model. Any changes or deviations of production cooling towers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish cooling towers conforming to this specification (see 3.2).

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the cooling tower covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91 (see 3.16.1).

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6.5 Registration numbers. The contracting officer should furnish a list of registration numbers to the contractor to be applied to each cooling tower (see 3.20.5).

6.6 Vehicle weight classification forms. The contracting officer should arrange to furnish the contractor with the applicable vehicle weight classification forms and instructions for completion. Further data may be obtained by contacting:

Commanding Officer
USA MERADCOM
ATTN: DRDME-HM
Fort Belvoir, VA 22060

6.7 Government-furnished property. When required, the contracting officer should arrange to furnish the property specified in 3.22.

6.8 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the cooling towers in accordance with this specification.

6.9 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each cooling tower.

6.10 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of the cooling tower.

6.11 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 3655-A084

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-C-52238C(ME) Cooling Tower, Liquid Semitrailer Mounted

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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